

INSTITUTE OF ACTUARIES OF INDIA

CS1 - Actuarial Statistics (Paper A)

November 2025 Examination

Indicative Solution

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

a) Calculate the probability that $X^2 > 9$

The condition $X^2 > 9$ is equivalent to $|X| > 3$, which means $X < -3$ or $X > 3$.

The total probability is the sum of two disjoint events:

$$P(X^2 > 9) = P(X < -3) + P(X > 3)$$

We standardize the X-values using the Z-score formula $Z = \frac{(X-5)}{2}$

$$\text{For } X < -3: \quad Z_1 = \frac{(-3-5)}{2} = -4, \quad P(X < -3) = P(Z < -4) \approx 0.00003167$$

$$\begin{aligned} \text{For } X > 3: \quad Z_2 &= \frac{(3-5)}{2} = -1, \quad P(X > 3) = P(Z > -1) \\ &= 1 - P(Z < -1) \approx 0.84134475 \end{aligned}$$

$$P(X^2 > 9) \approx 0.00003167 + 0.84134475 \approx 0.84137642 \approx 0.8414$$

[2 marks]

b) Profit P is normally distributed. The p_{th} percentile P_p is calculated as

$$P_p = \mu_p + z_p \sigma_p. \text{ We are given } \mu_p = 10 \text{ and } P_{80} = 16.$$

The Z-scores are:

$$z_{0.8} \approx 0.8416 \text{ and } z_{0.9} \approx 1.2816$$

$$16 = 10 + z_{0.8} \sigma_p$$

$$\sigma_p = \frac{6}{z_{0.8}} = \frac{6}{0.8416} \approx 7.1291$$

$$P_{0.9} = 10 + z_{0.9} \sigma_p = 10 + 1.2816 \times 7.1291 \approx 10 + 9.1363 \approx 19.14$$

[2 marks]

$$\text{c) } P(Y > 70) = P(e^X > 70) = P(X > \ln(70)); \ln(70) \approx 4.2485$$

$$P(X > 4.2485) = P(Z > \frac{4.2485 - 5}{2}) = P(Z > -0.37575)$$

$$P(Y > 70) = 1 - P(Z \leq -0.37575) \approx 1 - 0.35355 \approx 0.646$$

[2 marks]

[6 Marks]

Solution 2:

i) The Moment Generating Function (MGF) of a single Poisson random variable X with mean λ is defined as $M_X(t) = E[e^{tX}]$.

$$M_X(t) = \sum_{k=0}^{\infty} e^{tk} \frac{2^k e^{-2}}{k!} = e^{-2} e^{2e^t} = e^{2(e^t-1)} \text{ since } e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

Therefore, the answer is (b)

[2 marks]

ii)

$$M_S(t) = e^{30.2(e^t-1)} = e^{60(e^t-1)}.$$

Therefore, the answer is (b)

[2 marks]

iii) $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$. For a Poisson(λ) distribution: Mean: $\mu=\lambda=2$, Variance: $\sigma^2=\lambda=2$,

Sample size: $n=30$. $\bar{X} \sim N(2, \frac{1}{15})$.

$$Z = \frac{(\bar{X} - \mu)}{\sigma_{\bar{X}}} = \frac{2.2 - 2}{\sqrt{1/15}} \approx 0.7746, \text{ then}$$

$$P(\bar{X} < 2.2) \approx P(Z < 0.7746)$$

From the standard normal table:

$$\Phi(0.77) = 0.7794, \quad \Phi(0.78) = 0.7823$$

Using linear interpolation between 0.77 and 0.78:

$$\Phi(0.7746) = \Phi(0.77) + \frac{0.7746 - 0.77}{0.01} [\Phi(0.78) - \Phi(0.77)]$$

$$= 0.7794 + 0.46(0.7823 - 0.7794)$$

$$= 0.7794 + 0.46(0.0029)$$

$$= 0.7794 + 0.001334$$

$$= 0.7807$$

$$\boxed{P(\bar{X} < 2.2) = 0.7807}$$

[2 marks]

[6 Marks]

Solution 3:

i) $X \sim \text{Poisson}(\lambda = 2)$

$$\text{a) } E[X | X \geq 2] = \frac{\sum_{k=2}^{\infty} kP(X=k)}{P(X \geq 2)} = \frac{E[X] - \sum_{k=0}^1 kP(X=k)}{1 - P(X < 2)} = \frac{2 - 2e^{-2}}{1 - 3e^{-2}} \approx 2.911 \quad [2 \text{ marks}]$$

$$\text{b) } \text{Var}[X | X \geq 1] = E[X^2 | X \geq 1] - (E[X | X \geq 1])^2.$$

$$E[X | X \geq 1] = \frac{\sum_{k=1}^{\infty} kP(X=k)}{P(X \geq 1)} = \frac{E[X] - 0}{1 - P(X=0)} = \frac{2}{1 - e^{-2}}$$

$$E[X^2 | X \geq 1]: \text{First, } E[X^2] = \text{Var}[X] + (E[X])^2 = 2 + 2^2 = 6.$$

$$\sum_{k=1}^{\infty} k^2 P(X=k) = E[X^2] - 0^2 \cdot P(X=0) = 6$$

$$E[X^2 | X \geq 1] = \frac{6}{1 - P(X=0)} = \frac{6}{1 - e^{-2}}$$

$$\text{Var}[X | X \geq 1] = \frac{6}{1 - e^{-2}} - \left(\frac{2}{1 - e^{-2}} \right)^2 \approx 1.589 \quad [2 \text{ marks}]$$

ii) Let $N(t)$ be the number of alerts in t hours, $N(t) \sim \text{Poisson}(\lambda_t)$. Let T_4 be the time until the 4th alert.

The event that "the 4th alert arrives after more than one hour" ($T_4 > 1$) is the same as the event that "fewer than 4 alerts occur in the first hour" ($N(1) < 4$).

The number of alerts in 1 hour is $N(1) \sim \text{Poisson}(\lambda \cdot 1)$.

$$P(T4 > 1) = P(N(1) \leq 3) = \sum_{k=0}^3 \frac{\lambda^k e^{-\lambda}}{k!} = e^{-\lambda} (1 + \lambda + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!}) = 0.43 \quad [2 \text{ marks}]$$

iii)

Total arrival rate $\lambda_{total}=6$ per hour. Medium-priority (B) probability =0.45.

Find the rate for Medium-priority tickets: The tickets of a specific category form an independent Poisson process with rate:

$$\lambda_B = \lambda_{total} \cdot P(B) = 6 * 0.45 = 2.7 \text{ per hour}$$

Find the parameter for 2 hours: Let N_B be the number of B tickets in 2 hours.

$$N_B \sim \text{Poisson}(\lambda_B \cdot 2).$$

$$\lambda_{2h} = 2.7 * 2 = 5.4$$

$$P(N_B \geq 2):$$

$$P(N_B \geq 2) = 1 - P(N_B = 0) - P(N_B = 1)$$

$$P(N_B = 0) = e^{-5.4}$$

$$P(N_B = 1) = \frac{(5.4)^1 e^{-5.4}}{1!} = 5.4 e^{-5.4}$$

$$P(N_B \geq 2) = 1 - e^{-5.4} - 5.4 e^{-5.4} = 1 - 6.4 e^{-5.4} \approx 0.971 \quad [2 \text{ marks}]$$

iv)

Total arrival rate $\lambda_{total}=6$ per hour. A ticket is labeled "Low priority" if it arrives in the interval after 20 minutes (i.e., in the last 40 minutes of the hour).

Due to the independent increments property of the Poisson process, the number of tickets arriving in the last 40 minutes ($\frac{2}{3}$ hours) is independent of arrivals in the first 20 minutes.

Find the length of the Low-Priority interval:

$$t = 1 \text{ hour} - 20 \text{ min} = 40 \text{ minutes} = \frac{2}{3} \text{ hours}$$

Find the rate for the Low-Priority interval: Let N_{LP} be the number of Low-Priority tickets. $N_{LP} \sim \text{Poisson}(\lambda_{total} \cdot t)$.

$$\lambda_{LP} = 6 * \frac{2}{3} = 4$$

Calculate $P(N_{LP}=2)$:

$$P(N_{LP} = 2) = \frac{4^2 e^{-4}}{2!} = 8 e^{-4} \approx 0.1465 \quad [3 \text{ marks}]$$

[11 Marks]

Solution 4:

i)

Under $H_0: \theta = 4$, $X_i \sim \text{Gamma}(\alpha = 3, \theta = 4)^*$

$$\text{mean, } \mu_0 = 3 \times 4 = 12, \quad \text{variance, } \sigma_0^2 = 3 \times 4^2 = 48, \quad \text{So, } \sigma_0 = \sqrt{48}$$

Reject H_0 if $(\bar{X} \geq 14.5)$

Normal approximation: $\bar{X} \sim N\left(12, \frac{48}{n}\right)$ under H_0

Type I error probability – $P(\bar{X} \geq 14.5) \leq 0.04$

$$\Rightarrow \frac{14.5-12}{\sqrt{48/n}} \geq z_{0.04}, \text{ where } (z_{0.04} \approx 1.7507).$$

$$\frac{2.5}{\sqrt{48/n}} \geq 1.7507$$

$$\frac{6.25}{48/n} \geq 3.06495$$

$$n \geq \frac{3.06495 \times 48}{6.25} \approx 23.54$$

Minimum integer, $n = 24$

If any student makes an assumption $\lambda = 1/\theta$ and solves this problem first principles, 2 marks maximum may be given.

[3 marks]

ii)

Under H_0 , $\bar{X} \sim N\left(50, \frac{9}{10}\right)$, So, standard error, $\sigma_{\bar{X}} = \frac{3}{\sqrt{10}} \approx 0.948683$

Upper 5% critical value from standard normal: ($z_{0.05} = 1.64485$)

Reject H_0 if

$$\bar{X} \geq 50 + 1.64485 \times 0.948683 \approx 51.560$$

Under the true distribution, $\mu_1 = 52$, $\bar{X} \sim N\left(52, \frac{9}{10}\right)$.

Type II error probability, $\beta = P(\text{fail to reject } H_0 \mid \mu = 52)$

$$\beta = P(\bar{X} < 51.560 \mid \mu = 52)$$

$$Z = \frac{51.560 - 52}{0.948683} \approx -0.4637$$

$$\beta = \Phi(-0.4637) \approx 0.321$$

Thus, the probability of a Type II error is approximately 0.321.

[3 marks]

iii)

Given: $X \sim \text{Exp}(\theta)$,

test $H_0: \theta = 20$ vs $H_1: \theta = 30$, reject H_0 if $(X > c)$,

Type II error probability = 0.75.

$$\text{Under } (H_1): F_1(c) = 1 - e^{-c/30}. \text{ Type II error } \beta = F_1(c):$$

$$1 - e^{-c/30} = 0.75$$

$$e^{-c/30} = 0.25$$

$$c = 30 \cdot \ln(4) \approx 41.589$$

Type I error probability

$$\alpha = P(X > c \mid H_0) = e^{-c/20} \approx e^{-41.589/20} \approx e^{-2.079} \approx 0.125.$$

Therefore, $c \approx 41.59$ and $\alpha \approx 0.125$

[3 marks]

[9 Marks]

Solution 5:

i) Observed frequencies

	Under 25	Over 25
Like New	650	720
Like Old	150	180

Total sample size: $800 + 900 = 1700$

Row totals: Like New = $650 + 720 = 1370$, Like Old = $150 + 180 = 330$

Column totals: Under 25 = 800, Over 25 = 900

Expected frequencies under independence:

$$E_{ij} = \frac{\text{row total} \times \text{column total}}{\text{grand total}}$$

$$E_{\text{New, Under 25}} = \frac{1370 \times 800}{1700} = 644.7$$

$$E_{\text{New, Over 25}} = \frac{1370 \times 900}{1700} = 725.29$$

$$E_{\text{Old, Under 25}} = \frac{330 \times 800}{1700} = 155.29$$

$$E_{\text{Old, Over 25}} = \frac{330 \times 900}{1700} = 174.70$$

Test statistic:

$$\begin{aligned} \chi^2 &= \sum \frac{(O - E)^2}{E} \\ &= \frac{(650 - 644.7)^2}{644.7} + \frac{(720 - 725.29)^2}{725.29} + \frac{(150 - 155.29)^2}{155.29} + \frac{(180 - 174.70)^2}{174.70} \\ &= \frac{28.027}{644.7} + \frac{28.027}{725.29} + \frac{28.027}{155.29} + \frac{28.027}{174.70} \\ &= 0.0434 + 0.0386 + 0.18048 + 0.16042 \end{aligned}$$

$$= 0.4230$$

$$\text{Degrees of freedom: } (2 - 1)(2 - 1) = 1$$

$$\text{Critical value at 5\% significance: } \chi^2_{0.05}(1) = 3.841$$

Since $0.4230 < 3.841$, we fail to reject the null hypothesis of no association.

There is **no significant association** between age group and algorithm preference.

[5 marks]

ii) The behavioral model proposes the following population proportions:

$$\text{Under 25: Like New} = (1 - p)/2, \text{ Like Old} = p/2$$

$$\text{Over 25: Like New} = (1 - p^2)/2, \text{ Like Old} = p^2/2$$

$$\text{Maximum likelihood estimate: } p = 0.212$$

Expected proportions:

$$\text{Under 25 - Like New: } (1 - 0.212)/2 = 0.394$$

$$\text{Under 25 - Like Old: } 0.212/2 = 0.106$$

$$\text{Over 25 - Like New: } (1 - 0.212^2)/2 = 0.4775$$

$$\text{Over 25 - Like Old: } (0.212^2)/2 = 0.0225$$

$$\text{Check sum: Under 25 total} = 0.394 + 0.106 = 0.500,$$

$$\text{Over 25 total} = 0.4775 + 0.0225 = 0.500$$

Expected frequencies (sample size 800 per group):

$$\text{Under 25 - Like New: } 1700 \times 0.394 = 669.8$$

$$\text{Under 25 - Like Old: } 1700 \times 0.106 = 180.2$$

$$\text{Over 25 - Like New: } 1700 \times 0.4775 = 811.75$$

$$\text{Over 25 - Like Old: } 1700 \times 0.0225 = 38.2$$

Test statistic:

$$\begin{aligned} \chi^2 &= \sum \frac{(O - E)^2}{E} \\ &= \frac{(650 - 669.8)^2}{669.8} + \frac{(150 - 180.2)^2}{180.2} + \frac{(720 - 811.75)^2}{811.75} + \frac{(180 - 38.2)^2}{38.2} \\ &= \frac{392.04}{669.8} + \frac{912.04}{180.2} + \frac{8418.063}{811.75} + \frac{20093.06}{38.2} \\ &= 0.585 + 5.06 + 10.380 + 526.31 \\ &\approx 542.34 \end{aligned}$$

Degrees of freedom: 4 cells – 1 parameter – 1 = 2

Critical value at 5% significance: $\chi^2_{0.05}(2) = 5.991$

Since (542.34 > 5.991), we reject the null hypothesis that the model fits the data.

The behavioral model does not provide a good fit to the observed data. [5 marks]

[10 Marks]

Solution 6:

i)

Prior distribution: $p \sim \text{Beta}(\alpha_0 = 2, \beta_0 = 5)$

Likelihood: Binomial data with ($x = 60$) successes, ($n = 100$) trials

Posterior distribution: $p \mid \text{data} \sim \text{Beta}(\alpha_0 + x, \beta_0 + n - x)$

Substitute values: $\text{Beta}(2 + 60, 5 + 100 - 60) = \text{Beta}(62, 45)$

Correct answer: a) $\text{Beta}(62, 45)$ [2 marks]

ii)

Posterior mean formula: $\mu = \frac{\alpha}{\alpha + \beta}$

Calculation: $\mu = \frac{62}{62 + 45} = \frac{62}{107} \approx 0.5794$

Posterior variance formula: $\sigma^2 = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$

Calculation: $\sigma^2 = \frac{62 \times 45}{(107)^2 \times 108} = \frac{2790}{11449 \times 108} = \frac{2790}{1236492} \approx 0.002255$ [2 marks]

iii)

Standard deviation: $\sigma = \sqrt{0.002255} \approx 0.04749$

95% credible interval formula: $\mu \pm 1.96\sigma$

Calculation: $0.5794 \pm 1.96 \times 0.04749 = 0.5794 \pm 0.09308$

Interval: (0.4863, 0.6725)

Correct answer: b) (0.486, 0.672) [2 marks]

[6 Marks]

Solution 7:

i)

First calculate summary statistics:

$$n = 7, \quad \sum X = 21.5, \quad \sum Y = 145.1, \quad \sum XY = 489.85, \quad \sum X^2 = 74.75$$

$$\text{Mean of } X: \bar{X} = \frac{21.5}{7} \approx 3.0714$$

$$\text{Mean of } Y: \bar{Y} = \frac{145.1}{7} \approx 20.7286$$

Slope estimate:

$$\begin{aligned} \beta &= \frac{\sum XY - n\bar{X}\bar{Y}}{\sum X^2 - n\bar{X}^2} \\ &= \frac{489.85 - 7 \times 3.0714 \times 20.7286}{74.75 - 7 \times (3.0714)^2} \\ &= \frac{489.85 - 445.660}{74.75 - 66.034} = \frac{44.19}{8.716} \approx 5.07 \end{aligned}$$

Intercept estimate:

$$a = \bar{Y} - \beta\bar{X} = 20.7286 - 5.07 \times 3.0714 \approx 20.7286 - 12.703 \approx 5.156$$

$$\text{Fitted regression line: } \hat{y} = 5.156 + 5.07x \quad [3 \text{ marks}]$$

ii)

To test for a significant linear relationship, we test whether the slope coefficient is different from zero.

$$\text{Correct answer: a) } H_0: \beta = 0 \text{ vs } H_1: \beta \neq 0 \quad [1 \text{ mark}]$$

[4 Marks]

Solution 8:

i)

The Bernoulli distribution can be written in exponential family form:

$$f(y; p) = \exp \left[y \log \left(\frac{p}{1-p} \right) + \log(1-p) \right]$$

Compare with standard exponential family form:

$$f(y; \theta) = \exp \left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right]$$

From the Bernoulli distribution:

$$\text{--Canonical parameter: } \theta = \log \left(\frac{p}{1-p} \right)$$

$$\text{--} b(\theta) = \log(1 + e^\theta) \text{ since } p = \frac{e^\theta}{1 + e^\theta} \text{ and } 1 - p = \frac{1}{1 + e^\theta}$$

– Dispersion parameter: $\phi = 1$ for Bernoulli

– $a(\phi) = 1$

– $c(y, \phi) = 0$

Correct answer: b) $\theta = \log\left(\frac{p}{1-p}\right)$, $b(\theta) = \log(1 + e^\theta)$, $\phi = 1$, $a(\phi) = 1$, $c(y, \phi) = 0$

[2 marks]

ii)

The Gamma distribution can be written in exponential family form.

Let $\mu = k\beta$ be the mean.

Rewriting the PDF in exponential family form:

$$f(y; k, \beta) = \exp\left[-\frac{y}{\beta} + (k-1)\log y - k\log \beta - \log \Gamma(k)\right]$$

This can be rearranged as:

$$f(y; k, \beta) = \exp\left[\frac{y(-1/\beta) - (-\log(1/\beta))}{1/k} + (k-1)\log y - \log \Gamma(k)\right]$$

The canonical parameter is $\theta = -1/\beta$, and since $\mu = k\beta$,
we have $\beta = \mu/k$. So, $\theta = -k/\mu$

The canonical link function satisfies, $\theta = g(\mu)$

So, $g(\mu) = -k/\mu$

However, for the standard Gamma GLM (with known shape k), the canonical link is the reciprocal link ($1/\mu$), as the constant k can be absorbed into the linear predictor.

Correct answer: c) $g(\mu) = \frac{1}{\mu}$

[2 marks]

iii)

Analysis of Statement I: The saturated model has the maximum log-likelihood (lowest deviance), not the highest deviance. The null model (intercept only) typically has the highest deviance. Therefore, Statement I is false.

Analysis of Statement II: The deviance follows a χ^2 distribution asymptotically under certain regularity conditions, but this is not true for all models in the exponential family. For example, with binary data (Bernoulli distribution), the deviance does not follow a χ^2 distribution. Therefore, Statement II is false.

Both statements are incorrect.

Correct answer: e) None of the above

[2 marks]

[6 Marks]

Solution 9:

i) The slope estimate formula:

$$\widehat{\beta}_1 = \frac{\text{cov}(x, y)}{\text{var}(x)} = \frac{\text{cov}(x, y)}{\frac{1}{n-1} \sum (x_i - \bar{x})^2}$$

However, the given covariance (88.5) is already the unbiased sample covariance, which equals $\frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$

Therefore:

$$\widehat{\beta}_1 = \frac{\text{cov}(x, y) \times (n-1)}{\sum (x_i - \bar{x})^2} = \frac{88.5 \times 24}{1250} = \frac{2124}{1250} = 1.6992$$

Alternatively, using the relationship:

$$\widehat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{\text{cov}(x, y) \times (n-1)}{\sum (x_i - \bar{x})^2} = \frac{88.5 \times 24}{1250} = 1.6992$$

The intercept estimate:

$$\widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1 \bar{x} = 50 - 1.6992 \times 10 = 50 - 16.992 = 33.008$$

OLS estimates: $\widehat{\beta}_0 = 33.008$, $\widehat{\beta}_1 = 1.6992$ [2 marks]

ii) Calculate:

- a) Total degrees of freedom: $df_{\text{total}} = n - 1 = 20 - 1 = 19$
 Regression degrees of freedom: $df_{\text{regression}} = k = 1$ (one predictor)
 Error degrees of freedom: $df_{\text{error}} = n - k - 1 = 20 - 1 - 1 = 18$

[1 mark]

b) The regression sum of squares (SSR).

$$SSR = SST - SSE = 1450 - 522 = 928$$

[1 mark]

c) The coefficient of determination R^2

$$R^2 = \frac{SSR}{SST} = \frac{928}{1450} \approx 0.6393$$

[2 marks]

d) The value of the unbiased estimate of the error variance (σ^2) or MSE

$$MSE = \frac{SSE}{df_{\text{error}}} = \frac{522}{18} = 29$$

Un biased estimate of error variance: $\widehat{\sigma^2} = 29$

[2 marks]

iii)

a) Adjusted R^2 formula:

$$R_{\text{adj}}^2 = 1 - \left(\frac{n-1}{n-k-1} \right) (1 - R^2)$$

For Model – 1: $n = 50$, $k = 4$, $R^2 = 0.712$

$$R_{\text{adj}}^2 = 1 - \left(\frac{50-1}{50-4-1} \right) (1 - 0.712)$$

$$= 1 - \left(\frac{49}{45} \right) (0.288)$$

$$= 1 - (1.0889 \times 0.288)$$

$$= 1 - 0.3136 = 0.6864$$

Adjusted R^2 for Model – 1: 0.6864

[2 marks]

b) Limitations of (R^2):

- (R^2) always increases when more predictors are added, even if they are irrelevant
- It does not account for model complexity or overfitting
- It can lead to choosing models with too many parameters

Adjusted (R^2) addresses this by penalizing additional predictors:

$$R_{\text{adj}}^2 = 1 - \left(\frac{n-1}{n-k-1} \right) (1 - R^2)$$

For Model – 2: $n = 50$, $k = 7$, $R^2 = 0.725$

$$R_{\text{adj}}^2 = 1 - \left(\frac{49}{50-7-1} \right) (1 - 0.725)$$

$$= 1 - \left(\frac{49}{42} \right) (0.275)$$

$$= 1 - (1.1667 \times 0.275)$$

$$= 1 - 0.3208 = 0.6792$$

Comparison:

–Model – 1 Adjusted(R^2): 0.6864

–Model – 2 Adjusted(R^2): 0.6792

Although Model 2 has a higher R^2 (0.725 vs 0.712), the Adjusted (R^2) for Model 2 is lower than for Model 1 (0.6792 vs 0.6864). This demonstrates that the three additional predictors in Model 2 do not provide meaningful explanatory power and may be overfitting the data. The

analyst's statement is incorrect because (R^2) alone is not a reliable metric for comparing models with different numbers of predictors. [3 marks]

iv)

a)

Gamma GLM with log link: $E[Y] = e^\eta$.

Given coefficients:

$$\eta = 12.62 + (-0.25) \times R + 0.4 \times H$$

$$\text{Urban House: } (\eta = 12.62 + 0.4 \Rightarrow \text{Price} = e^{13.02} \approx 451,350)$$

$$\text{Rural Apartment: } (\eta = 12.6 - 0.25 = 12.37 \Rightarrow \text{Price} = e^{12.37} \approx 235,625).$$

Thus answer is (C).

[2 marks]

b)

Given predictions:

	Urban	Rural
House	300,000	260,000
Apartment	447,540	420,000

Model with interaction:

$$\log(\mu) = \beta_0 + \beta_1 R + \beta_2 H + \beta_3 (R \times H)$$

where $(R = 1)$ for Rural, (0) for Urban; $(H = 1)$ for House, (0) for Apartment.

From the data:

$$\ln(447540) = \widehat{\beta}_0$$

$$\ln(420000) = \widehat{\beta}_0 + \widehat{\beta}_1$$

$$\ln(300000) = \widehat{\beta}_0 + \widehat{\beta}_2$$

$$\ln(260000) = \widehat{\beta}_0 + \widehat{\beta}_1 + \widehat{\beta}_2 + \widehat{\beta}_3$$

Solving these equations:

$$\widehat{\beta}_0 = \ln(447540) \approx 13.011$$

$$\widehat{\beta}_1 = \ln(420000) - \widehat{\beta}_0 \approx -0.063$$

$$\widehat{\beta}_2 = \ln(300000) - \widehat{\beta}_0 \approx -0.400$$

$$\widehat{\beta}_3 = \ln(260000) - (\widehat{\beta}_0 + \widehat{\beta}_1 + \widehat{\beta}_2) \approx -0.080$$

Thus, the estimated beta for the interaction of Rural and House is (-0.080) . [3 marks]

[18 Marks]

Solution 10:

i) Variance of PC3 = (Standard deviation of PC3)² = $(1.2214)^2 = 1.4918$ [1 mark]

ii)

1. Since the components chosen are uncorrelated linear combinations of the variables of the data, this excludes the possibility of variables being correlated to each other.
2. **Dimensionality Reduction / Noise Reduction:** It reduces the number of input features, which can help prevent overfitting in predictive models, reduce computational cost, and potentially improve model performance by filtering out noise. [2 marks]

iii) Kaiser's criterion: Retain components with variance > 1

- PC1 variance = $(1.8741)^2 = 3.512 > 1$
- PC2 variance = $(1.6327)^2 = 2.666 > 1$
- PC3 variance = $(1.2214)^2 = 1.492 > 1$
- PC4 variance = $(0.9452)^2 = 0.893 < 1$
- Therefore, 3 components should be retained. [2 marks]

iv) Spearman's rank correlation formula:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

Calculate the squared differences:

$$\sum d^2 = 1^2 + 0^2 + (-2)^2 + 0^2 + (-2)^2 + 2^2 + (-1)^2 + (-2)^2$$

$$\sum d^2 = 1 + 0 + 4 + 0 + 4 + 4 + 1 + 4 = 18$$

With $n = 8$:

$$\rho = 1 - \frac{6 \times 18}{8 \times (64 - 1)} = 1 - \frac{108}{8 \times 63} = 1 - \frac{108}{504} = 1 - 0.2143 = 0.7857$$

Spearman's rank correlation coefficient: $\rho \approx 0.786$

This indicates a strong positive correlation between customers' spending ranks in Month 1 and Month 3.

[4 marks]

[9 Marks]**Solution 11:****i)***Prior distribution: $\theta \sim \text{Beta}(\alpha, \beta)$* *Likelihood for one batch: $X_i \sim \text{Binomial}(m, \theta)$* *Total number of defective items across all batches: $\sum_{i=1}^n X_i$* *Total number of components across all batches: $n \times m$* *Total number of non – defective items: $nm - \sum_{i=1}^n X_i$* *The prior distribution for (θ) has the form:*

$$\text{Prior}(\theta) \propto \theta^{\beta-1}(1-\theta)^{\alpha-1}$$

*The random variable (X) has a binomial distribution with parameters (m) and (θ) .**So the likelihood function based on a random sample $(x_1, \dots, x_n)$ is:*

$$L(\theta) = \binom{m}{x_1} \theta^{x_1} (1-\theta)^{m-x_1} \times \binom{m}{x_2} \theta^{x_2} (1-\theta)^{m-x_2} \times \dots \times \binom{m}{x_n} \theta^{x_n} (1-\theta)^{m-x_n}$$

$$\propto \theta^{\sum x_i} (1-\theta)^{mn - \sum x_i}$$

The posterior distribution for (θ) is proportional to the product of the likelihood function and the prior distribution.

$$\text{Posterior}(\theta) \propto \theta^{\beta-1}(1-\theta)^{\alpha-1} \times \theta^{\sum x_i} (1-\theta)^{mn - \sum x_i}$$

$$= \theta^{\beta-1+\sum x_i} (1-\theta)^{\alpha-1+mn-\sum x_i}$$

Correct answer: a) $\text{Beta}(\alpha + \sum_{i=1}^n X_i, \beta + nm - \sum_{i=1}^n X_i)$

[2 marks]

ii)*We have already worked out the likelihood function based on the sample data:*

$$L(\theta) = C \theta^{\sum x_i} (1-\theta)^{mn - \sum x_i}$$

Taking logs:

$$\log L = K + \sum X_i \log \theta + (mn - \sum X_i) \log(1 - \theta)$$

Differentiating with respect to θ :

$$\frac{d}{d\theta} \log L = \frac{\sum X_i}{\theta} - \frac{mn - \sum X_i}{1 - \theta}$$

Setting this expression equal to zero and rearranging, we get the MLE for θ :

$$\hat{\theta} = \frac{\sum x_i}{mn}$$

We can check that this is a maximum by differentiating the log likelihood a second time:

$$\frac{d^2}{d\theta^2} \log L = -\frac{\sum x_i}{\theta^2} - \frac{mn - \sum x_i}{(1 - \theta)^2}$$

Since $mn - \sum x_i$, must be positive, this expression is negative,

and the estimate we have found is indeed a maximum.

Therefore, we get the MLE for θ :

$$\hat{\theta} = \frac{\sum x_i}{mn}$$

Correct answer is (a)

[2 marks]

iii)

The posterior distribution is beta distribution with parameters $(\beta + \sum x_i)$ and $(\beta + mn - \sum x_i)$ (from part (i)(a)). So, the Bayesian estimate for (θ) is using the formula for the mean of the beta distribution given in the Tables):

$$\hat{\theta} = \frac{\beta + \sum x_i}{\beta + \sum x_i + \beta + mn - \sum x_i} = \frac{\beta + \sum x_i}{2\beta + mn}$$

We need to rewrite this as a credibility estimate:

$$Zg(x) + (1 - Z)$$

where (μ) is the mean of our prior distribution, so $\mu = \frac{1}{2}$

We can split this up as follows:

$$\begin{aligned} \frac{\beta + \sum x_i}{2\beta + mn} &= \frac{\sum x_i}{2\beta + mn} + \frac{\beta}{2\beta + mn} \\ &= \frac{mn}{2\beta + mn} \times \frac{\sum x_i}{mn} + \frac{2\beta}{2\beta + mn} \times \frac{1}{2} \\ &= Z \times \frac{\sum x_i}{mn} + (1 - Z) \times \frac{1}{2} \end{aligned}$$

$$\text{where } Z = \frac{mn}{2\beta + mn}$$

Therefore, the correct answer is (a)

[2 marks]

iv)

(d) Each production batch contains ($m = 8$) components, and observations from ($n = 5$) batches are as follows:

$$x_1 = 2, \quad x_2 = 1, \quad x_3 = 0, \quad x_4 = 3, \quad x_5 = 2$$

Calculate the Bayesian estimate of the defect probability θ and the corresponding credibility factor (Z) when ($\beta = 6$)

Let's first calculate ($\sum x_i$)

$$\sum x_i = 2 + 1 + 0 + 3 + 2 = 8$$

Now, let's calculate Bayesian estimate of (θ)

Using the formula from part (i)(c):

$$\hat{\theta} = \frac{\beta + \sum x_i}{2\beta + mn}$$

$$\hat{\theta} = \frac{6 + 8}{2 \times 6 + 8 \times 5} = \frac{14}{12 + 40} = \frac{14}{52} = \frac{7}{26} \approx 0.2692$$

Calculate credibility factor (Z)

$$Z = \frac{mn}{2\beta + mn}$$

$$Z = \frac{8 \times 5}{2 \times 6 + 8 \times 5} = \frac{40}{12 + 40} = \frac{40}{52} = \frac{10}{13} \approx 0.7692$$

Verification using credibility form

$$\hat{\theta} = Z \times \frac{\sum x_i}{mn} + (1 - Z) \times \frac{1}{2}$$

$$\hat{\theta} = \frac{10}{13} \times \frac{8}{40} + \left(1 - \frac{10}{13}\right) \times \frac{1}{2}$$

$$\hat{\theta} = \frac{10}{13} \times \frac{1}{5} + \frac{3}{13} \times \frac{1}{2}$$

$$\hat{\theta} = \frac{10}{65} + \frac{3}{26} = \frac{2}{13} + \frac{3}{26} = \frac{4}{26} + \frac{3}{26} = \frac{7}{26} \approx 0.2692$$

Final Answer:

$\hat{\theta} = \frac{7}{26} \approx 0.2692, \quad Z = \frac{10}{13} \approx 0.7692$
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The Bayesian estimate suggests that approximately 26.92% of components are defective, with a credibility factor of about 76.92%, indicating that the data has substantial influence on the estimate compared to the prior.

[3 marks]

[9 Marks]**Solution 12:****i)**

The Neyman – Pearson lemma states that

if C is a critical region of size α and there exists a constant k such that $\frac{L_0}{L_1} \leq k$ inside C and $\frac{L_0}{L_1} \geq k$ outside C ,

then C is a most powerful critical region of size α
for testing the simple hypothesis $\theta = \theta_0$ against the simple alternative hypothesis $\theta = \theta_1$

[2 marks]

ii)

Type I error: Rejecting the null hypothesis (H_0) when it is actually true

Type II error: Failing to reject the null hypothesis (H_0) when the alternative hypothesis (H_1) is true.

[2 marks]

iii)

The size of a test is the probability of committing a Type I error, denoted by α . It is the maximum probability of rejecting (H_0) when (H_0) is true, defined as $\alpha = \sup_{\theta \in \Theta_0} P(\text{reject } H_0 | \theta)$

[1 mark]

iv)

The power of a test is the probability of correctly rejecting the null hypothesis when the alternative hypothesis is true, denoted by $(1 - \beta)$ where (β) is the probability of Type II error. It is defined as $\text{Power}(\theta) = P(\text{reject } H_0 | \theta)$ for $(\theta \in \Theta_1)$.

[1 mark]

[6 Marks]
